Name_____ Date_____ Period_____

LESSON 9-2 · HOMEWORK PACKET · SUPPORTED · SESSION 1 · PAGES 1–2

From Table to Graph

Maryland Standard 6.EE.C.9 · Learning target – I can turn a table of values into points and plot them on a graph.

REMEMBER

- Each row of a table becomes an **ordered pair** (x, y).
- To plot (x, y), start at (0, 0). Move **right** x, then **up** y.
- Points from a constant rate line up in a **straight line**.

WORKED EXAMPLE

Movie tickets cost \$8 each. Write ordered pairs for 1, 2, and 3 tickets.

Step 1 Tickets is x. Cost is y. Multiply each x by 8.

Step 2 $8 \times 1 = 8,\ 8 \times 2 = 16,\ 8 \times 3 = 24$

Answer **(1, 8), (2, 16), (3, 24)**

WORD BANK

ordered pair – (x, y): x first, then y
x-coordinate – how far right
y-coordinate – how far up
plot – mark a point on a graph

PRACTICE

show your work · label units

1. Write each table row as an ordered pair (x, y).

Step 1 Write the x value first.

Step 2 Write the y value second, inside ( ).

a. $x = 4, y = 12$ _____

b. $x = 0, y = 0$ _____

c. $x = 6, y = 30$ _____

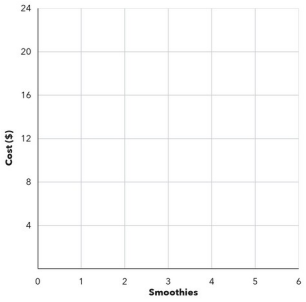
d. $x = 9, y = 2$ _____

2. Smoothies. Smoothies cost \$4 each. Complete the table. Then plot the points on the grid and connect them.

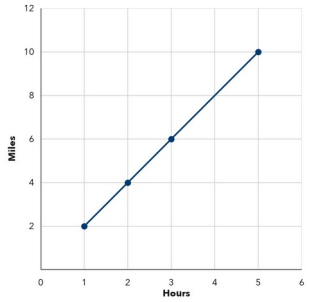
Step 1 Multiply each number of smoothies by \$4.

Step 2 Plot each pair: go right x, then up y.

Smoothies (x)	1	2	3	5
Cost in \$ (y)				



3. Hiking. The graph shows how far Kofi hikes.



a. Write the ordered pair for 5 hours. _____

b. What does (3, 6) mean? _____

c. Miles per hour: _____

4. Bike Rental. A bike costs **$6** for each hour. Write the ordered pairs (hours, cost) for 1, 3, and 4 hours. Then tell what (0, 0) means.

Step 1 Multiply each number of hours by 6.

Step 2 Write (hours, cost).

Ordered pairs_____

5. Find the Error. Teo's table shows *x* (hours): 2, 4 and *y* (dollars): 10, 20. He plotted (10, 2) and (20, 4). Explain his mistake. Which points should he plot?

Frame *Teo's mistake is ___. He should plot ___ and ___.*

_____

_____

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___

AT HOME Show someone at home how to plot (3, 5): right 3, then up 5.

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LESSON 9-2 · HOMEWORK PACKET · SUPPORTED · SESSION 2 · PAGES 3–4

Analyzing a Graph

Maryland Standard 6.EE.C.9 · Learning target – I can read a relationship off a graph and describe the rate it shows.

REMEMBER

- For a line through (0, 0), the **unit rate** is $y \div x$. The point (3, 12) means 3 of x go with 12 of y.
- With the same scales, a **steeper** line has a greater rate.

WORKED EXAMPLE

A line through (0, 0) passes through (2, 10). Find the unit rate and y when $x = 7$.

Step 1 Unit rate: $10 \div 2 = 5$

Step 2 $y = 5 \times 7 = 35$

Answer **5 per unit; $y = 35$**

WORD BANK

unit rate – the amount for 1 of x

steeper – rises faster

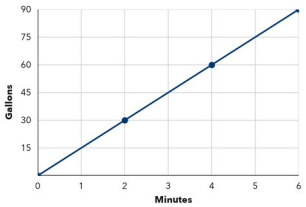
origin – the point (0, 0)

proportional – a straight line through (0, 0)

PRACTICE

show your work · label units

1. Filling a Pool. The graph shows the water in a pool.



Step 1 Find (4, 60) on the graph. Read the labels on both axes.

Step 2 For 1 minute, divide the gallons by the minutes.

a. What does (4, 60) mean? _____

b. Gallons per minute: _____

c. Gallons after 9 minutes: _____

2. Each line passes through (0, 0) and the given point. Find the unit rate.

Step 1 Divide y by x.

a. (3, 21) _____ **b.** (5, 40) _____ **c.** (8, 96) _____

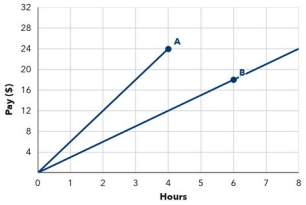
3. Reading. A line through (0, 0) shows that Ana reads 18 pages in 3 hours. Find her unit rate. How many pages does she read in 7 hours?

Step 1 Divide the pages by the hours to find pages in 1 hour.

Step 2 Multiply that by 7.

Answer_____

4. Two Jobs. The graph shows pay for two jobs. Find each pay rate. Which line is steeper, and what does it mean?



Answer_____

5. Savings. Jada's savings graph passes through (0, 10), (1, 15), and (2, 20). She says it is proportional. Is she right? What does (0, 10) mean?

Frame *Jada is ___ because a proportional graph ___. (0, 10) means ___.*

TEST PRACTICE

choose the best answer

6. A line through the origin passes through (5, 35). What is the unit rate?

A 5 **B** 30 **C** 7 **D** 175

7. Line P passes through (1, 4). Line Q passes through (1, 9). Both start at (0, 0). Which is true?

A P is steeper.
B Q is steeper; greater rate.
C They have the same rate.
D Q is steeper; smaller rate.

CHECK YOURSELF	☐ I can do this on my own	☐ I need a little help	☐ I have a question about #___